BUSINESS PROBLEM STATEMENT (Iris Dataset)

We need to analyze the **Iris dataset** to gain insights into the characteristics of different **iris flower species** based on their measurable features — *sepal length, sepal width, petal length, and petal width.*
The goal is to build a **predictive classification model** that can accurately identify the **species of iris flower** (Setosa, Versicolor, or Virginica) based on these measurements.

This analysis will help in understanding the relationships among various flower dimensions and enable automated identification of species for **botanical research and agricultural applications**.

KPI'S REQUIREMENT

We want to compute and visualize the following **key indicators and metrics** from the Iris dataset to understand data patterns and model performance:

1. **TOTAL RECORDS:**
   The total number of flower samples available in the dataset.
2. **SPECIES DISTRIBUTION:**
   The count and percentage of samples belonging to each iris species.
3. **AVERAGE FEATURE VALUES BY SPECIES:**
   Mean sepal length, sepal width, petal length, and petal width for each species.
4. **FEATURE CORRELATION MATRIX:**
   To identify how features relate to each other (for example, correlation between petal length and petal width).
5. **OUTLIER DETECTION SUMMARY:**
   Count and percentage of potential outliers across each numerical feature.

CHARTS REQUIREMENT

We would like to visualize the following aspects of the Iris dataset to better understand relationships and patterns:

1. **SPECIES DISTRIBUTION:**
   Create a pie chart or bar chart showing the number of samples per iris species.
2. **FEATURE COMPARISON BY SPECIES:**
   Generate boxplots for each feature (sepal length, sepal width, petal length, petal width) grouped by species to analyze variation.

3. **PAIR PLOT (SCATTER MATRIX):**
   Create a scatter matrix to visualize the pairwise relationships among all numeric features colored by species.
4. **CORRELATION HEATMAP:**
   Visualize correlations among all numeric features to identify the most influential relationships.